

A proposal for a type trait to detect narrowing conversions

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Project: Programming Language C++

Audience: Study Group 6 (Numerics), Library Evolution Working Group Incubator

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Introduction

This paper proposes a new type trait for the C++ Standard Library, `is_narrowing_convertible`, to detect whether a type is implicitly convertible to another type by going through a narrowing conversion.

Changelog

- P0870R2
 - Incorporated feedback from the LEWGI, EWGI, EWG sessions in Prague. In the LEWGI session this proposal was met favourably (2-10-2-1-0). EWGI and EWG didn't have any direct feedback (and also no particular concerns).
 - Elaborated on possible implementations.
 - Elaborated more on design decisions, especially regarding whether user-defined types should be handled, and whether this proposal would constrain core language evolution. Unlike R1, we now also include user-defined types in narrow conversion sequences (but like R1, we stick to `[dcl.init.list]` semantics, which are not customizable).
 - Updated reference to P0608R3.
 - Referenced other papers.
 - Incorporated the changes to `[dcl.init.list]` introduced by P1957R2.
 - Added SG6 to the target audience.
 - Improved spelling.
- P0870R1
 - Submitted for the Prague mailing, targeting LEWGI.

Motivation and Scope

A *narrowing conversion* is formally defined by the C++ Standard in `[dcl.init.list]`, with the intent of forbidding them from list-initializations.

It is useful in certain contexts to know whether a conversion between two types would qualify as a narrowing conversion. For instance, it may be useful to inhibit (via SFINAE) construction or conversion of "wrapper" types (like `std::variant` or `std::optional`) when a narrowing conversion is requested.

A use case the author has recently worked on was to prohibit narrowing conversions when establishing a connection in the Qt framework (see [\[Qt\]](#)), with the intent of making the type system more robust. Simplifying, a connection is established between a *signal* non-static member function of an object and a *slot* non-static member function of another object. The signal's and the slot's argument lists must have compatible arguments for the connection to be established: the signal must have at least as many arguments as the slot, and the each argument of the signal must be implicitly convertible to the corresponding argument of the slot. In case of a mismatch, a compile-time error is generated. Bugs have been observed when connections were successfully established, but a narrowing conversion (and subsequent loss of data/precision) was happening. Detecting such narrowing conversions, and making the connection fail for such cases, would have prevented the bugs.

Impact On The Standard

This proposal is a pure library extension.

It proposes changes to an existing header, `<type_traits>`, but it does not require changes to any standard classes or functions and it does not require changes to any of the standard requirement tables.

This proposal does not require any changes in the core language, and it has been implemented in standard C++.

This proposal does not depend on any other library extensions.

This proposal may help with the implementation of [\[P0608R3\]](#), the adopted resolution for [\[LEWG 227\]](#). In particular, in [\[P0608R3\]](#)'s wording section, the "invented variable `x`" is used to detect a narrowing conversion. However, the same paragraph is also broader than the current proposal, in the sense that it special-cases boolean conversions `[bool.conv]`, while the current proposal does not handle them in any special way. The part about boolean conversions has been anyhow removed by adopting [\[P1957R2\]](#).

The just mentioned [P1957R2], as a follow up of [P0608R3], made conversions from pointers to `bool` narrowing. It was adopted in the Prague meeting, therefore extending the definition of narrowing conversion in [dcl.init.list]. We do not see this as a problem: we aim to provide a trait to detect narrowing conversions exactly as specified by the Standard. Should the specification change, then the trait detection should also change accordingly; cf. the design decisions below.

This proposal overlaps with [P1818R1], a proposal that aims at introducing a distinction between narrowing and widening conversions. [P1818R1] is much more ambitious than this proposal; amongst other things, it aims at defining a widening trait. The wording of [P1818R1] is not complete, but we think that such a trait might simply yield the negation of the trait proposed here; it would ultimately depend on the exact definition of what constitutes a "widening conversion". Anyway, the adoption of [P1818R1] is orthogonal to the present proposal.

[P1619R1] and [P1998R1] introduce functions (called respectively `can_convert` and `is_value_lossless_convertible`) that check whether a given value of an integer type can be represented by another integer type. This is in line with the spirit of detecting narrowing conversions, namely, preventing loss of information and/or preventing undefined behavior. While this proposal works on types, the functions examine specific values; we therefore think that the proposals are somehow orthogonal to the current proposal. Moreover, [P1619R1] and [P1998R1]'s functions only deal with integer types, while the narrowing definition in [dcl.init.list] (adopted by this proposal) also deals with floating point and enumeration types (and, following [P1957R2]'s adoption, also boolean and pointer types).

Design Decisions

The most natural place to add the trait presented by this proposal is the already existing `<type_traits>` header, which collects most (if not all) of the type traits available from the Standard Library.

In the `<type_traits>` header the `is_convertible` type trait checks whether a type is implicitly convertible to another type. Building upon this established name, the proposed name for the trait described by this proposal shall therefore be `is_narrowing_convertible`, with the corresponding `is_narrowing_convertible_v` type trait helper variable template.

Should `is_convertible_v<From, To> == true` be a precondition of trying to instantiate `is_narrowing_convertible<From, To>`?

The author deems this unnecessary. Adding such a precondition would likely make the usage of the proposed trait more difficult and/or error prone.

As far as the author can see, the precondition is not going to dramatically simplify the specification for the proposed trait, or its implementation.

In terms of usage: such precondition would force the usage of `std::conjunction` (or similar short-circuit behavior) rather than a simple logical AND to detect whether there is a non-narrowing conversion between two datatypes, which is what the proposed type trait will be used for most of the times. In other words, we claim that the most common use case will be an expression like `is_convertible_v<From, To> && !is_narrowing_convertible_v<From, To>`. Therefore, we believe that it will be convenient for end users to avoid the pitfalls related to making `is_convertible_v` a precondition.

Should `is_narrowing_convertible_v<From, To>` yield `true` only for the combinations of types listed in [dcl.init.list]? What about user-defined types?

The R1 revision of this paper was limiting the possibility for the trait to yield `true` if and only if both `From` and `To` were one of the types listed in [dcl.init.list], which are all builtin types. Now, while we strongly want the type trait to match the language definition of narrowing conversion, there are some objections to simply detecting if we are in one of the those cases.

For instance, should a conversion from `const double` to `float` be considered narrowing or not? If we take the listing in [dcl.init.list] literally, then the answer would be "no" (that is, the trait would have to yield `false`), because `const double` is not in the list. However, this is not really useful, nor expected. If some code checks whether there is a conversion between two types and the conversion isn't narrowing, such code could erroneously infer that this is the case between `const double` and `float` and do the conversion. (This is in spite of the fact that list-initialization would not work, because *there is* a narrowing conversion in the conversion sequence).

To elaborate on this last sentence: the definition of list-initialization in [dcl.init.list] contains several cases where, if a narrowing conversion takes place, then the program is ill-formed. We believe that, more than detecting if a given conversion between two types *is* a narrowing conversion (as strictly per the definition), users want to detect if a conversion *has* a narrowing conversion, and thus would be forbidden if used in a list-initialization. As a consequence, we claim that `is_narrowing_convertible_v<const double, float>` should yield `true`.

This is not simply doable by stripping `From` and `To` of their *cv*-qualifiers (e.g. by applying `std::decay_t` on them), and then checking if the resulting types form a narrowing conversion. One must also consider all the other cases for conversions, which necessarily include user-defined types, as they may define implicit conversion operators that can be used as part of a conversion sequence that also includes a narrowing conversion. As an example:

```
struct S { operator double() const; }
S s;
float f1 = s;      // OK
float f2 = { s }; // ERROR: narrowing
```

We therefore claim that `is_narrowing_convertible_v<S, float>` should yield `true`. A similar example is used in [\[P0608R3\]](#) and then [\[P1957R2\]](#)'s design. Slightly adapted:

```
struct Bad { operator char const*() &&; };
```

With the adoption of [\[P1957R2\]](#) in the Standard, we claim that `is_narrowing_convertible_v<Bad, bool>` should yield `true`. For completeness, `is_narrowing_convertible_v<Bad &, bool>` should yield `false` (there is no conversion), and `is_narrowing_convertible_v<Bad &&, bool>` should yield `true`.

Note that, although user-defined datatypes are involved, the detection of whether there is a narrowing conversion sticks to the list of narrowing conversions in `[dcl.list.init]`, which deals with fixed number of cases: between certain integer and floating point types; from unscoped enumeration types to a integer or a floating point type; and from pointer types to boolean. We are not proposing to allow any customization to these cases (see below).

Should `is_narrowing_convertible_v<From, To>` yield false for boolean conversions `[conv.bool]`?

The author deems this to be out of scope for the proposed trait, which should have only the semantics defined by the language regarding narrowing conversion. Another trait should be therefore added to detect (implicit) boolean conversions, which, according to the current wording of `[dcl.init.list]`, are not considered narrowing conversions. (Note that with the adoption of [\[P1957R2\]](#), conversions from pointer types to boolean are indeed considered narrowing.)

It is the author's opinion that, in general, implicit boolean conversions are undesirable for the same reasons that make narrowing conversions unwanted in certain contexts — in the sense that a conversion towards `bool` may discard important information. However, we recognize the importance of not imbuing a type trait with extra, not yet well-defined semantics, and therefore we are excluding boolean conversions from the present proposal.

Should `is_narrowing_convertible<From, To>` use some other semantics than the narrowing defined in `[dcl.init.list]`?

Given that narrowing is formally defined in the core language, we are strongly against deviating from that definition. If anything, proposals aiming at marking as narrowing certain conversions between user defined types should also amend the core language definition; the type trait would still stick to that definition.

Should specializations of `is_narrowing_convertible<From, To>` be allowed?

We believe that specializations of the trait should be forbidden. This is consistent with the other type traits defined in `[meta]` (cf. `[meta.rqmts]`), and with the fact that the current wording of `[dcl.init.list]` defining narrowing conversions does not extend to user-defined types.

We are aware of broader work happening in the area, such as [\[P1818R1\]](#). That proposal introduces a keyword to mark widening conversions, but one may think of an alternative approach where users are expected to specialize `is_narrowing_convertible` in order to mark a conversion as narrowing. In any case, lifting the above limitation on specializations would always be possible.

Does providing such a trait make changes to the definition of "narrowing conversion" harder to land?

This objection has been raised during the LEWGI meeting in Prague. The objection is sound, in the sense that a future possible change to the definition of narrowing conversion will impact user code using the trait.

On the other hand, during the Prague meeting, EWGI and EWG did not raise this concern regarding this proposal.

First and foremost, we would like to remark that the trait is already implementable — and has indeed already been implemented — using standard C++ (e.g. via `To{std::declval<From>() }`, cf. below). Any change to the definition of narrowing would therefore already be impacting user code. The impact would likely go beyond type traits / metaprogramming, since initialization itself would change semantics. An example of such a change affecting `[dcl.init.list]` comes from [\[P1957R2\]](#), which has been adopted in Prague. That adoption changed the semantics of code like `bool b = {ptr};` (making it ill-formed; was OK before).

In general, changes to the core language that affected type traits have already happened. For instance, between C++17 and C++20 the definition of "aggregate" in `[dcl.init.aggr]` has been changed, therefore changing the semantics of the `std::is_aggregate` type trait. Similarly, the definitions of "trivial class" and of "trivially copyable class" in `[class.prop]` have changed between C++14 and C++17; meaning that also the type traits `std::is_trivial` and `std::is_trivially_copy_constructible` have changed semantics.

We can therefore reasonably assume that the presence of a trait does not seem to preclude changes to the core language (or, if such an objection was raised when proposing these changes to the core language, it was also disregarded, as the changes did ultimately land).

Finally, it is the author's opinion that code that is using facilities to detect narrowing conversions would like to stick to the core language definition. If the definition changes, we foresee that the detection wants to change too, following the definition. Note that an ad-hoc solution (rather than a standardized trait) may fall out of sync with the core language definition, and thus start misdetecting some cases.

Should there be a matching concept?

This point was raised during the LEWGI meeting in Prague, and it was deemed unnecessary at this point, because ultimately this proposal is just about a type trait. If it will be deemed useful, adding a concept could always be done at a later time, via another proposal.

Bikeshedding: `is_narrowing_convertible` or `is_non_narrowing_convertible`?

The Standard Library does not currently have type traits with a negation in their names, so the current proposal uses a positive tense, although we foresee that the majority of the usages are going to be about detecting whether there is *no* narrowing conversion between two given types. Note that Standard Library names such as `is_nothrow_constructible` are not considered to be negations; the tense is positive, using `nothrow` as an adjective/attribute.

Bikeshedding: `is_narrowing_convertible` or `is_narrowing`?

This paper proposes the former, building on top of the existing `is_convertible` name; however, we concede that `is_narrowing` could lead to cleaner code. Moreover, at the present time, there should be no ambiguity over the meaning of `is_narrowing`: the only usage of "narrowing" in the Standard is about narrowing conversions. Nonetheless, it may make sense to future-proof the name proposed here, and therefore go for the slightly more verbose `is_narrowing_convertible`.

Technical Specifications

The proposed type trait has already been implemented in various codebases using ad-hoc solutions (depending on the quality of the compiler, etc.), using standard C++ and without the need of any special compiler hooks.

One possible implementation is to exhaustively check whether a conversion between two types fall in one of the narrowing cases listed in [dcl.list.init]. This particular implementation has been done in [\[Qt.connect\(\)\]](#); it has obviously maintenance and complexity shortcomings. The most important part is:

```
template<typename From, typename To, typename Enable = void>
struct AreArgumentsNarrowedBase : std::false_type
{
};

template<typename From, typename To>
struct AreArgumentsNarrowedBase<From, To, typename std::enable_if<sizeof(From) && sizeof(To)>::type>
    : std::integral_constant<bool,
        (std::is_floating_point<From>::value && std::is_integral<To>::value) ||
        (std::is_floating_point<From>::value && std::is_floating_point<To>::value && sizeof(From) > sizeof(To)) ||
        ((std::is_integral<From>::value || std::is_enum<From>::value) && std::is_floating_point<To>::value) ||
        (std::is_integral<From>::value && std::is_integral<To>::value
         && (sizeof(From) > sizeof(To)
          || (std::is_signed<From>::value ? !std::is_signed<To>::value
            : (std::is_signed<To>::value && sizeof(From) == sizeof(To)))))) ||
        (std::is_enum<From>::value && std::is_integral<To>::value
         && (sizeof(From) > sizeof(To)
          || (IsEnumUnderlyingTypeSigned<From>::value ? !std::is_signed<To>::value
            : (std::is_signed<To>::value && sizeof(From) == sizeof(To))))))
    >
{
};
```

(Historical note: at the time this implementation was done, the codebase could only use C++11 features, and some compilers were unable to cope with the SFINAE solution presented below.).

The Qt implementation strictly checks for the one of narrowing cases listed in [dcl.init.list], which is not the aim of the current proposal (cf. design decisions above). Also, the snippet does not yet implement the change introduced by [\[P1957R2\]](#), showing an inherent limitation to such an "exhaustive" solution: it may fall out of sync with the core language.

Another, much simpler implementation is to rely on SFINAE around an expression such as `To{std::declval<From>() }`. However, this form is also error-prone: it may accidentally select aggregate initialization for `To`, or a constructor taking a `std::initializer_list`, and so on. If for any reason these make the expression ill-formed, then the trait is going to report that there is a narrowing conversion between the types, even when there actually isn't one (or vice-versa, depending on how one builds the trait).

[\[P0608R3\]](#) uses a clever workaround to avoid falling into these traps: declaring an array, that is, building an expression like `To t[] = { std::declval<From>() };`. This solution is now used in the Standard (in `[variant.ctor]` and `[variant.assign]`), amongst other things, in order to detect and prevent a narrowing conversion. The trait that we are proposing could be used to replace the ad-hoc solution, and clarify the intent of it.

Another possible implementation, that uses the just mentioned workaround, is the following one (courtesy of Zhihao Yuan; received via private communication):

```

template<class From, class To>
inline constexpr bool is_narrowing_convertible_v = false;

template<class T, class U>
concept __construct_without_narrowing = requires (U&& x) {
    { std::type_identity_t<T[]>{std::forward<U>(x)} } -> T[1];
};

template<class From, class To> requires std::is_convertible_v<From, To>
inline constexpr bool is_narrowing_convertible_v<From, To> =
    !__construct_without_narrowing<To, From>;

```

We claim that there is an inherent danger here: it is extremely likely that non-experts would go for a "naïve", wrong solution (e.g. `To{std::declval<From>()})`). This further underlines the necessity of providing detection of narrowing conversions in the Standard Library, rather than having users reinventing it with ad-hoc solutions.

Standardese

In [meta.type.synop], add to the the header synopsis:

```

template <class From, class To> struct is_convertible;
template <class From, class To> struct is_narrowing_convertible;

```

```

template <class From, class To>
inline constexpr bool is_convertible_v = is_convertible<From, To>::value;
template <class From, class To>
inline constexpr bool is_narrowing_convertible_v = is_narrowing_convertible<From, To>::value;

```

In [meta.rel], add a new row to the *Type relationship predicates* table, in the same row as for `is_convertible`:

Template template <class From, class To> struct is_convertible; template <class From, class To> struct is_narrowing_convertible;

Condition see below

Comments From and To shall be complete types, cv void, or arrays of unknown bound.

Modify paragraph 5 of [meta.rel]:

5. The predicate condition for a template specialization `is_convertible<From, To>` or a template specialization `is_narrowing_convertible<From, To>` shall be satisfied if and only if the return expression in the following code would be well-formed, including any implicit conversions to the return type of the function; for `is_narrowing_convertible<From, To>` the conversions shall, moreover, require a narrowing conversion ([dcl.init.list]):

```

To test() {
    return declval<From>();
}

```

[*Note*: This requirement gives well-defined results for reference types, void types, array types, and function types. — *end note*]
[*Note*: For the purpose of establishing whether the conversion is a narrowing conversion, the source is never a constant expression. — *end note*]

Acknowledgements

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References

[Qt] Qt version 5.15, [*QT_NO_NARROWING_CONVERSIONS_IN_CONNECT*](#), QObject class documentation

[Qt connect ()] Qt version 5.15, [*traits for detecting narrowing conversions*](#)

[P0608R3] Zhihao Yuan, [*A sane variant converting constructor \(LEWG 227\)*](#).

[LEWG 227] Zhihao Yuan, [*Bug 227 - variant converting constructor allows unintended conversions*](#)

[P1957R2] Zhihao Yuan, [*Converting from \$T^*\$ to bool should be considered narrowing \(re: US 212\)*](#)

[P1818R1] Lawrence Crowl, [*Narrowing and Widening Conversions*](#)

[P1619R1] Lisa Lippincott, [*Functions for Testing Boundary Conditions on Integer Operations*](#)

[P1998R1] Ryan McDougall, [*Simple Facility for Lossless Integer Conversion*](#)